MILITARY SPECIFICATION

OPENER, CAN, HAND, FOLDING

This specification is mandatory for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a lightweight, folding, hand-operated can opener for severing tops of rimmed metal cans.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-700	-	Steel Sheet and Strip, Medium and High Carbon
PPP-B-636	-	Box, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing)

MILITARY

MIL-T-10727	-	Tin Plating; Electrodeposited or Hot-Dipped, For Ferrous or Nonferrous Metals

FSC 7330

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

DRAWING

ARMY NATICK LABORATORIES

2-9-148 - Opener, Can, Hand, Folding

(Miniature copy of Drawing 2-9-148, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Specification E18-67 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Applications for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.

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3. REQUIREMENTS

* 3.1 First article.- This specification contains provisions for first article test and approval (see 4.2, 6.2 and 6.4).

3.2 Materials.-

3.2.1 Carbon steel, sheet and strip.- Carbon steel sheet and strip shall conform to number 1050, 1055, or 1065 of QQ-S-700.

3.3 Design and construction.- The can opener shall be designed and constructed as specified herein and as shown on Drawing 2-9-148. The opener shall completely sever the tops of circular and rectangular cans when tested as specified in 4.4.3. The opener shall be assembled so the knife blade offers resistance to changing from the closed position to the open position and from the open position to the closed position. Opening of the blade shall be accomplished by use of the hand without the use of an auxiliary instrument. It is not required that the blade be retained firmly in the open position.

3.3.1 Heat treatment.- The body and the blade shall be heat treated to a Rockwell "C" through hardness of not less than 47 nor more than 55 and shall be verified by the test specified in 4.4.2.

3.3.2 Knife blade.- The knife blade shall have a sharp, tapered, cutting edge formed by grinding, milling, shearing, or coining.

3.3.3 Finish.- All burrs and rough edges shall be removed. The opener shall be tin plated in accordance with type I of MIL-T-10727. The thickness of plating on all surfaces shall be not less than 0.0001 inch and shall be verified by the test specified in 4.4.1.

3.4 Identification marking.- Each can opener shall be permanently and distinctly stamped with the letters "U.S." and the manufacturer's name or trademark. Markings shall be of the size and in the location shown on Drawing 2-9-148. Markings shall be applied so there is no distortion or unevenness opposite the markings or penetration through the metal.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- a preproduction sample is required, it shall be examined for the defects listed in table I and dimensions specified and tested in accordance with 4.4.1, 4.4.2, and 4.4.3. Failure to pass any test shall be cause for rejection of the preproduction sample. The presence of any defects listed in table I or any dimension not within specified requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Inspection of components and materials.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Inspection of the end item.- The inspection lot shall consist of all can openers offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated can opener.

4.3.2.1 Visual examination.- The end item shall be examined for defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of visual defects

Examine	Defect	Classification Major	Minor
Finish	Any burr or rough edge	X	
	Tin plating omitted, area of no coating, or rust	X	
	Base metal shows through plating	X	
	Plating not adherent (flaky, peeling, or blistered)	X	
	Plating scratched, stained, or discolored		X

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TABLE I.- Classification of visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Blade and body (general)	Design not as shown on drawing	X	
	Opener not assembled as specified	X	
	Any component missing, broken, cracked, malformed, or bent out of shape	X	
	Corners not rounded where required		X
Blade	No cutting edge	X	
	Cutting edge not tapered level or not uniform		X
	Blade cannot be opened to the operating position by hand without the use of an auxiliary instrument	X	
	Blade offers no resistance to changing from closed to open or from open to closed		X
Body	Hole missing	X	
	Body not embossed (reinforcing rib)		X
Identification marking	Missing, illegible, incomplete, incorrect, or not in proper location		X
	Penetration on opposite surface	X	
Diagram and directions on envelope	Missing, illegible, or incorrect	X	
	Envelope torn or mutilated		X

4.3.2.2 Dimensional examination.- Dimensional examination shall be made of the completed can openers to determine compliance with Drawing 2-9-148 and as specified herein. Any nonconformance shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.3.2.3 End item testing.- The blade and body shall be tested for plating thickness and hardness in accordance with 4.4.1 and 4.4.2 respectively, using inspection level S-3 with an AQL of 2.5 expressed in terms of defects

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per hundred units. The can opener shall be tested as specified in 4.4.3 using inspection level S-2 with an AQL of 1.0 expressed in terms of defects per hundred units.

4.3.3 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table II. The lot shall be the number of shipping containers offered for inspection at one time. The sample unit shall be one container fully prepared for delivery. The inspection level shall be S-3 with an AQL of 6.5, expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Examine	Defect
Markings (exterior and interior container)	Omitted, incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged or otherwise defective.
Contents (exterior and interior container)	Number per container is more or less than required. Count less than specified or indicated quantity in intermediate box.

* 4.3.3.1 Examination of shipping containers.- Examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

* 4.3.4 Examination of palletization.- An examination shall be made to determine that palletization is in compliance with section 5 of this specification. Defects shall be scored as specified in table III. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

TABLE III.- Examination of palletization

Examination	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.

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TABLE III.- Examination of palletization (cont'd)

Examination	Defect
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.4 Tests.-

4.4.1 Plating thickness.- Plating thickness shall be tested by the nondestructive method as specified in MIL-T-10727 to determine conformance with 3.3.3. Any nonconformance with the specified requirements shall constitute failure of this test.

* 4.4.2 Hardness.- Hardness shall be tested in accordance with ASTM E18-67 to determine conformance with 3.3.1. Any nonconformance with the specified requirements shall constitute failure of this test.

* 4.4.3 Cutting.- Each sample can opener shall be tested to determine that it completely severs tops of both circular and rectangular cans as required by 3.3 The test shall be performed on one commercial No. 2 circular can and one rectangular can measuring approximately 6-5/8 inches long by 4-1/8 inches wide by a minimum of 1-5/8 inches high. Any nonconformance with the specified requirements shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Unit packaging.- Each can opener, with the blade folded flat, shall be packaged in an envelope made from material comprised of not less than 0.0005-inch thick polyethylene coated on not less than 25 pounds basis weight (24-by 36-500) unbleached kraft paper and the seams shall be formed by heat sealing. The envelope containing one can opener shall be closed by heat sealing. Unless otherwise specified (see 6.2), the maximum dimensions of the envelope shall be 1-11/16 inches by 2-3/4 inches.

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5.1.1.2 Intermediate packaging.- One thousand can openers, unit packaged as specified in 5.1.1.1, shall be random-packaged in a snug-fitting fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. Each box shall be closed by covering all seams full length with strips of 2-1/2-inch minimum width tape conforming to type III, grade B of PPP-T-45 or PPP-T-76. The center strip shall extend a minimum of 2-1/2-inches onto the end panels. Alternatively, the bottom flaps may be closed in accordance with method II of the appendix of PPP-B-636.

* 5.1.2 Level C.- Can openers, unit packaged as specified in 5.1.1.1, shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Four thousand can openers, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding, shall be in accordance with the appendix of PPP-B-636.

* 5.2.2 Level B.- Four thousand can openers, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

* 5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

* 5.2.3 Level C.- Can openers packaged as specified in 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Palletization.- Unless otherwise specified (see 6.2), can openers packed as specified shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be

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in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, unit packages, intermediate packages, shipping containers, and unit loads shall be marked in accordance with MIL-STD-129.

5.4.1 Special marking.- Each unit package shall have printed thereon a diagram and directions for use as shown on Drawing 2-9-148.

6. NOTES

6.1 Intended use.- The can opener covered herein is intended for use by military personnel in the field to open combat-type rations, in-flight rations, and emergency rations.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article (see 3.1, 4.2, and 6.4).
(c) Envelope dimensions, if other than specified (see 5.1.1.1).
(d) Selection of the applicable level of packaging and packing (see 5. and 5.2).
(e) When weather-resistant grade fiberboard is required for level B pack (see 5.2.2.1).
* (f) When palletization is not required (see 5.3).

6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

* 6.4 First article.- The item covered by this specification requires first article test and approval uner the appropriate provisions of paragraph 7-104.5 of the Armed Services Procurement Regulations. The first article should be a preproduction sample and should consist of one unit. The contracting office should include specific instructions in all procurement instruments, regarding arrangements for examination test, and approval of the first article.

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Custodians:

Army - GL
Navy - SA
Air Force - 84

Review activities:

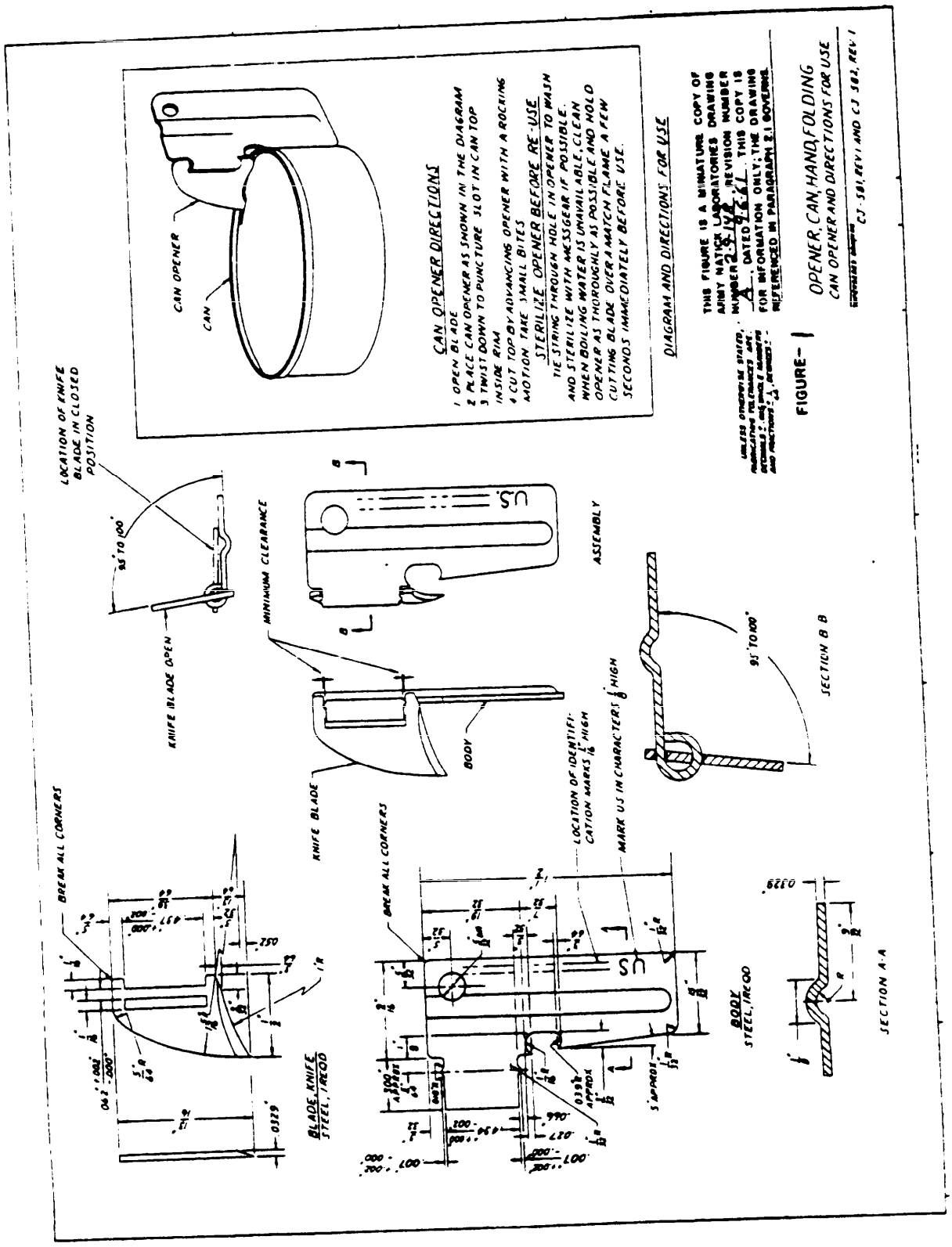
Army - MD
Navy - MC

Preparing activity:

Army - GL

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requesting activity believes it to be. In the event of a conflict between the

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-O-20582D - OPENER, CAN, HAND, FOLDING

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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